



INTRODUCTION TO MED 101 FOUNDATIONAL CONCEPTS IN HUMAN ANATOMY

REFERENCES:

- *Netter Atlas of Human Anatomy*, 8th Ed. Section 1 (Introduction).
- *Snell Clinical Anatomy*, 4th Ed. Chapter 1 (Thorax) and Chapter 2 (Abdomen).
- *BRS Cell Biology & Histology*, 6th Ed. Chapter 9 (Nervous System)

OUTLINE

I. Anatomical Terminology

- Standard Anatomical Position
- Directional Terms
- Regional Terms
- Anatomical Movements

II. Body Planes, Regions, and Cavities

- Anatomical Planes
- Body Regions
- Body Cavities and Membranes

III. Levels of Structural Organization

- Structural Organization
- Relationship Between Structure and Function
- Homeostasis and Body Organization

IV. Overview of Organ Systems

- Integumentary System
- Skeletal System
- Muscular System
- Nervous System
- Endocrine System
- Cardiovascular System
- Lymphatic and Immune System
- Respiratory System
- Digestive System
- Urinary System
- Reproductive System

I. ANATOMICAL TERMINOLOGY

STANDARD ANATOMICAL POSITION

- Universal reference point for all anatomical descriptions.
- The body is **erect**, facing forward, with the head and toes directed anteriorly. The arms are at the sides with the **palms facing forward (anteriorly)**

DIRECTIONAL TERMS

Superior (Cranial/Rostral) vs. Inferior (Caudal): Superior is toward the top of the skull; inferior is toward the soles of the feet (or tail eminence in embryos)

Anterior (Ventral) vs. Posterior (Dorsal): Anterior describes the front of the body or limbs; posterior describes the back

Medial vs. Lateral: Medial refers to structures closer to the median plane (midline); lateral refers to those farther away

Proximal vs. Distal: Used for limbs; proximal is closer to the trunk attachment, while distal is farther from the point of attachment

Superficial vs. Deep: Terms describing position relative to the surface of the body

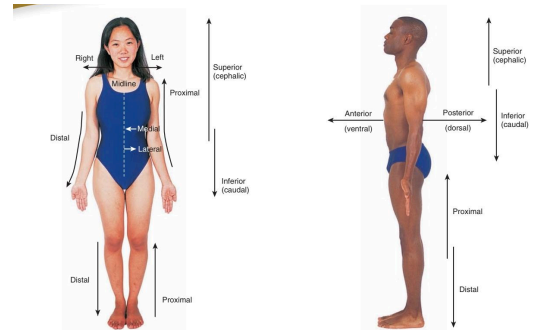


FIGURE 1.9 Directional Terms
All directional terms are in relation to the body in the anatomical position: a person standing erect with the face directed forward, the arms hanging to the sides, and the palms of the hands facing forward. **APR**

REGIONAL TERMS

Axial Division: Head, neck, and trunk

- **Head and Neck:** Includes the frontal (forehead), orbital (eyes), nasal (nose), oral (mouth), and mental (chin) regions
- **Trunk:** Composed of the thorax (chest), abdomen, and perineum. Posterior landmarks include the vertebral (midline), scapular (shoulder blade), and lumbar (lower back) regions

Appendicular Division: Upper and lower limbs (extremities)

- **Limbs:** The upper limb includes the shoulder, axilla (armpit), arm (brachium), elbow, forearm (antebrachium), wrist, and hand. The lower limb includes the hip, thigh (femoral), knee, leg, ankle, and foot

Major Regional Subdivisions:

- *Cephalic (Head), Cervical (Neck), Thoracic (Chest), Abdominal, Pelvic, Perineal*
- *Upper Limb: Axillary, Brachial, Antebrachial, Carpal, Palmar*
- *Lower Limb: Femoral, Patellar, Popliteal, Crural, Tarsal, Plantar*

ANATOMICAL MOVEMENTS

Flexion/Extension: Decreasing or increasing the angle of a joint (e.g., at the elbow or knee).



SCHOOL OF MEDICINE

MED101: HUMAN ANATOMY AND HISTOLOGY

BATCH 2030

Abduction/Adduction: Moving a limb away from or toward the midline.

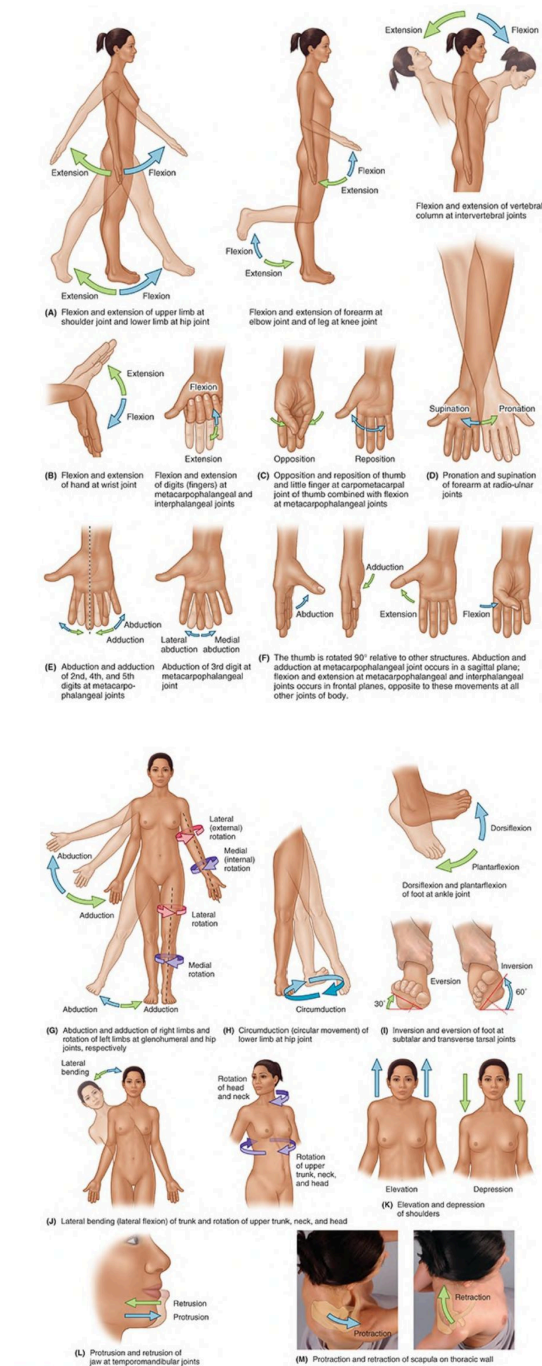
Rotation: Medial (internal) or lateral (external) turning of a bone around its longitudinal axis.

Specialized:

Pronation/Supination (forearm)

Inversion/Eversion (foot)

Dorsiflexion/Plantarflexion (ankle)



II. BODY PLANES, REGIONS, AND CAVITIES

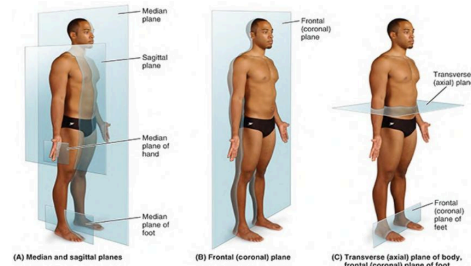
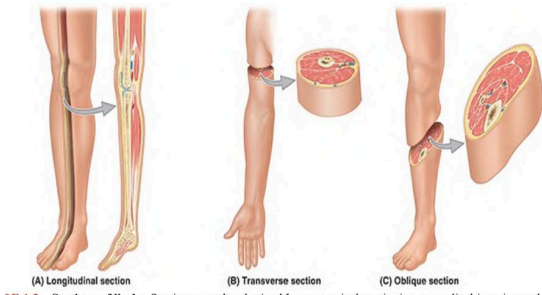


FIGURE 1.2. Anatomical position and anatomical planes. The main planes of the body.



Sagittal Planes:

- **Midsagittal (Median):** Divides body into equal left and right halves
- **Parasagittal:** Divides body into unequal left and right halves

Frontal (Coronal) Plane: Divides body into anterior (front) and posterior (back) portions

Transverse (Horizontal / Axial) Plane: Divides body into superior (top) and inferior (bottom) portions

Oblique Plane: Passes through the body at an angle between transverse and sagittal/frontal planes

BODY CAVITIES AND MEMBRANES

Dorsal (Posterior) Body Cavity:

- **Cranial Cavity:** Encloses the brain
- **Vertebral (Spinal) Cavity:** Encloses the spinal cord

Ventral (Anterior) Body Cavity:

- **Thoracic Cavity:**
 - **Pleural Cavities:** Enclose the lungs
 - **Pericardial Cavity:** Encloses the heart (within the mediastinum)



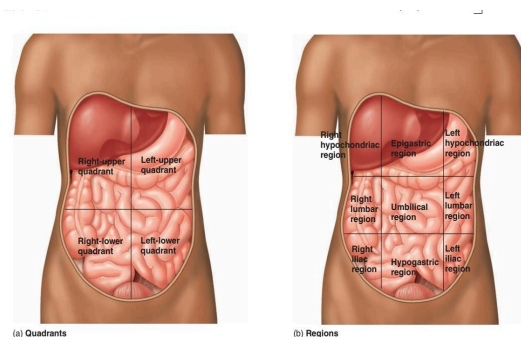
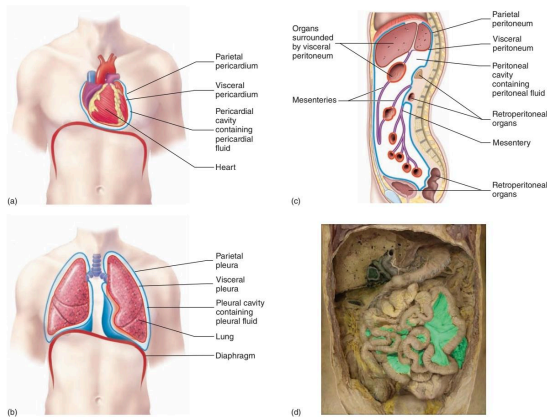
- **Mediastinum:** Central space containing heart, esophagus, trachea, and major blood vessels
- **Abdominopelvic Cavity:**
 - **Abdominal Cavity:** Contains stomach, intestines, liver, spleen, gallbladder
 - **Pelvic Cavity:** Contains urinary bladder, reproductive organs, rectum

Abdominopelvic Subdivisions:

- **4 Quadrants:** RUQ, LUQ, RLQ, LLQ
- **9 Regions:** Epigastric, Umbilical, Hypogastric, Right/Left Hypochondriac, Right/Left Lumbar, Right/Left Iliac (Inguinal)

Serosus Membranes (Serosa):

- Parietal layer (lines cavity wall) vs. Visceral layer (covers organs)
- Pleura, Pericardium, and Peritoneum



III. LEVELS OF STRUCTURAL ORGANIZATION

Atomic and molecular building blocks such as DNA, RNA, and proteins

Atoms: The smallest stable units of matter (e.g., carbon, hydrogen, oxygen, nitrogen, calcium, potassium).

Molecules: Combinations of two or more bound atoms forming simple inorganic compounds (such as water, which comprises 70%–80% of cellular mass, and dissolved electrolytes) or complex organic macromolecules (lipids, carbohydrates, proteins, and nucleic acids).

Biological Relevance: Chemical interactions govern cellular metabolism, membrane potentials, and genetic encoding.

CELLULAR LEVEL

The basic structural and functional unit of life; animal cells are eukaryotic with a distinct nucleus and membranous organelles

Cells: The basic structural and functional units of living organisms.

Protoplasm & Subcellular Components: Composed of water, electrolytes (K^+ , Mg^{2+} , PO_4^{3-} , HCO_3^-), functional/structural proteins, lipids (phospholipids, cholesterol), and glycogen.

Organelles: Specialized intracellular structures—such as the nucleus (genetic control), endoplasmic reticulum (protein and lipid synthesis), Golgi apparatus (packaging and glycosylation), lysosomes (hydrolytic digestion), peroxisomes (fatty acid β -oxidation), and mitochondria (aerobic ATP generation)—that isolate specific biochemical reactions.

TISSUE LEVEL

Groups of similar cells and extracellular matrix (ECM) performing specific functions

All organs are composed of four basic types: Epithelial, Connective, Muscular, and Nervous

Tissues: Groups of similar cells and their surrounding intercellular matrix performing specific, shared functions.

The Four Primary Tissue Types:

1. **Epithelial Tissue:** Covers body surfaces, lines hollow organs and cavities, and forms glandular structures. Provides protection, absorption, and secretion.

2. **Connective Tissue:** Supports, binds, and protects organs while storing energy and providing structural framework (e.g., bone, blood, adipose, cartilage).



3. Muscle Tissue: Produces physical movement, posture maintenance, and heat through active contraction (Skeletal, Cardiac, and Smooth muscle).

4. Nervous Tissue: Detects environmental changes, processes sensory input, and transmits electrical impulses for rapid internal communication.

ORGAN LEVEL

Different tissues integrated to perform specific tasks, such as the heart, stomach, or liver

Functional cells are called parenchyma, and supporting connective tissue is called stroma

Organs: Anatomical structures composed of two or more primary tissue types combined in definitive patterns to perform specific physiological tasks.

Structural Integration: For example, the stomach combines epithelial lining (secretion and absorption), smooth muscle layers (mechanical churning), connective tissue layers (vascular and structural support), and neural tissue (autonomic innervation and motility control).

ORGAN SYSTEM LEVEL

A group of organs (e.g., the digestive system) working together for a common purpose

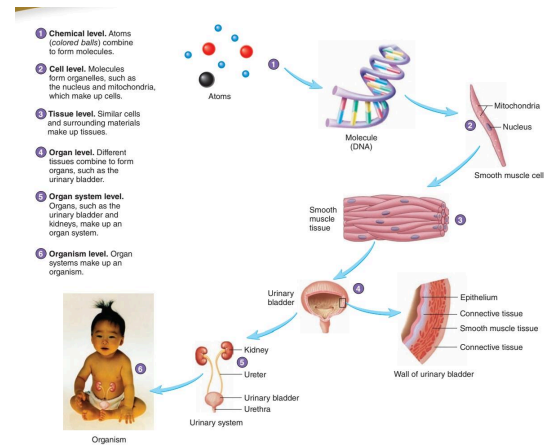
Organ Systems: Groups of related organs that cooperate to achieve broad homeostatic goals.

System Interdependence: The human body consists of 11 major organ systems (Integumentary, Skeletal, Muscular, Nervous, Endocrine, Cardiovascular, Lymphatic, Respiratory, Digestive, Urinary, and Reproductive). No single system functions in isolation; each relies on the metabolic and structural output of the others.

ORGANISMAL LEVEL

The sum total of all structural levels working together to maintain the human body

The Integrated Human Body: The highest level of structural organization, representing the total sum of all structural levels working in coordinated harmony to maintain life and health.



RELATIONSHIP BETWEEN STRUCTURE AND FUNCTION

1. Principle of Complementarity of Structure and Function

Core Concept: Function (physiology) is inherently dependent upon structure (anatomy). What an organ or cell can do is strictly constrained by its physical design and molecular composition.

Bi-directional Influence: Evolutionary and developmental forces adapt anatomical structures to optimize physiological processes, while mechanical and chemical forces continuously remodel tissue architecture in response to physiological demand.

2. Form-Function Relationships Across Levels of Organization

Molecular & Cellular Level

- The amphipathic structure of phospholipids (hydrophilic phosphate heads and hydrophobic fatty acid tails) spontaneously forms a selective permeability barrier, preventing the free movement of water-soluble substances into cells.

- Microtubules in a 9 + 2 axoneme pattern, powered by ATP-dependent dynein arms, physically slide past one another to generate the mechanical whipping action of cilia.

Tissue & Microvascular Level

- Alveoli in the lungs consist of ultra-thin, simple squamous epithelium adjacent to pulmonary capillary endothelium, minimizing diffusion distance for rapid gas exchange (O_2 and CO_2).

- Renal tubular epithelial cells are packed with mitochondria and express high concentrations of active transport pumps ($Na^+/K^+-ATPase$) to supply



the massive energy demand required for active solute reabsorption.

Organ & System Level

- The thick, muscular walls of the left ventricle generate high hydraulic pressure to drive systemic circulation, whereas the thinner right ventricle pumps blood under low pressure to the pulmonary circuit.
- The vast surface area of the intestinal mucosa—expanded by plicae circulares, villi, and microvilli—maximizes nutrient absorption into circulating blood and lymph.

HOMEOSTASIS AND BODY ORGANIZATION

1. The Internal Environment (Milieu Intérieur)

Definition: Claude Bernard's concept of the milieu intérieur refers to the Extracellular Fluid (ECF), which bathes all body cells and serves as the bridge between the external world and cellular metabolism.

Fluid Compartment Division

Intracellular Fluid (ICF): Constitutes approximately 2/3 of total body water; rich in K^+ , Mg^{2+} , and PO_4^{3-} .

Extracellular Fluid (ECF): Constitutes approximately 1/3 of total body water; rich in Na^+ , Cl^- , and HCO_3^- . Includes interstitial fluid and blood plasma.

Homeostatic Balance: Homeostasis is the active maintenance of nearly constant physical and chemical conditions within the ECF (e.g., pH 7.35–7.45, glucose 75–95 mg/dL, arterial $PO_2 \approx 40$ mmHg).

2. Homeostatic Control Systems

Control loops regulate biological parameters within tight physiological ranges through specific reflex components:

Stimulus → Receptor (Sensor) → Control Center → Effector → Response → Restores Homeostasis

Negative Feedback Mechanisms

- The primary homeostatic control mechanism. Detects a deviation from a set point and triggers a response in the opposite direction to restore baseline.
- Example: Elevated arterial pressure stretches carotid and aortic baroreceptors → signals sent to medullary vasomotor centers → ↓ sympathetic

outflow → arterial vasodilation and decreased cardiac output → blood pressure returns to normal.

- $\text{Gain} = \text{Correction} \div \text{Uncorrected Error}$

Positive Feedback Mechanisms

- Amplifies the original stimulus, driving the variable further from baseline. Requires a definitive external mechanism to terminate the cycle; otherwise, it leads to instability.
- Physiological Examples: Blood coagulation cascade, uterine contractions during parturition via oxytocin release, and rapid phase-0 depolarization of action potentials via voltage-gated Na^+ channels.

- Pathological Example: Hemorrhagic shock, where acute blood loss reduces cardiac output, compromising coronary artery perfusion, weakening cardiac muscle, further reducing cardiac output, and leading to death.

Adaptive (Feed-Forward) Control

- Anticipatory control used primarily by the central nervous system (e.g., cerebellum) for rapid motor execution. Corrective signals are applied to subsequent motor attempts after analyzing delayed sensory feedback.

3. Organ System Integration in Homeostasis

Nutrient and Gas Supply

- Respiratory System: Captures O_2 and excretes cellular CO_2 .
- Digestive System: Breaks down nutrients and absorbs carbohydrates, fatty acids, and amino acids into the ECF.

- Cardiovascular System: Circulates ECF, rapidly mixing interstitial fluid and blood plasma across capillaries.

Metabolic Waste Removal

- Urinary System: Filters arterial blood to remove metabolic end-products (urea, uric acid, excess ions) and regulates fluid/acid-base status.
- Liver: Metabolizes toxins, converts excess ammonia (NH_3) to urea, and excretes bile pigments.

Systemic Regulation & Control

- Nervous System: Controls rapid responses via sensory input, central integration, and motor/autonomic output.



- **Endocrine System:** Regulates long-term metabolic, ionic, and growth functions via hormonal secretion into the bloodstream.

4. Homeostatic Imbalance

Pathophysiology: Disease is defined as a state of disrupted homeostasis. When one or more body systems fail to maintain ECF parameters within normal non-lethal limits, cellular dysfunction occurs.

Pathological Vicious Cycles: Prolonged failure of negative feedback systems can cause positive feedback loops to take over, leading to tissue damage, multi-organ failure, or clinical death.

IV. OVERVIEW OF ORGAN SYSTEMS

INTEGUMENTARY SYSTEM

- **Main Functions:** Acts as a major barrier against the environment, provides thermoregulation via sweat and blood flow, and serves as a primary sensory organ.
- **Major Components:** Composed of the **epidermis** (epithelial layer) and the **dermis** (connective tissue layer), plus appendages like hair, nails, and glands.
- **Clinical Importance:** Heaviest single organ (approx. 16% of body weight); vital for assessing hydration, circulation, and systemic disease.

Gross Anatomy

- **Location/Surface Anatomy:** Covers the entire external body surface.
- **Parts:** **Thick skin** (palms/soles) and **thin skin** (everywhere else).
- **Blood Supply:** Supplied by the **subpapillary plexus** at the dermal-epidermal junction and the **subcutaneous plexus**.
- **Innervation:** Richly innervated by cutaneous nerves with specialized receptors like **Meissner corpuscles** (touch) and **Pacinian corpuscles** (vibration).

Embryology

- **Embryonic Origin:** Epidermis arises from **surface ectoderm**; dermis arises from **mesoderm**.
- **Development:** Starts as a single layer of ectoderm at 4 weeks, becoming multilayered by week 11.
- **Congenital Anomalies:** Includes **angiomas** (vascular tumors) and **ectodermal dysplasia**.

Histology and Microscopic Anatomy

- **Tissue Organization:** Stratified squamous keratinized epithelium (epidermis) over dense irregular connective tissue (dermis).
- **Cell Types:** **Keratinocytes** (predominant), **Melanocytes** (pigment), **Langerhans cells** (immune), and **Merkel cells** (touch).
- **Layers:** **Stratum basale** (mitosis), **spinosum** (desmosomes), **granulosum** (keratohyalin), **lucidum** (thick skin only), and **corneum** (dead keratinized squames).
- **Specialized Structures:** **Arrector pili muscles** (smooth muscle for hair) and **sebaceous/sweat glands**.

Clinical and Imaging Anatomy

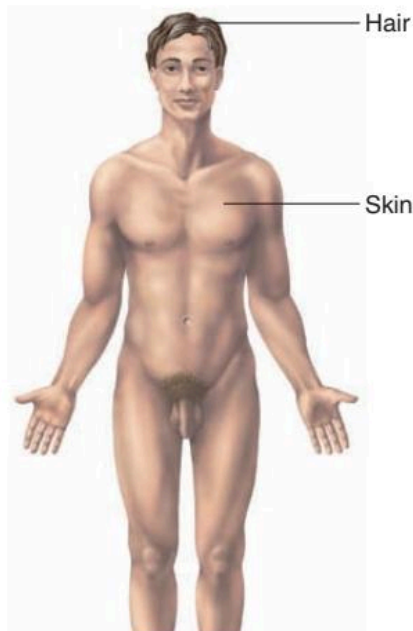
- **Common Diseases:** **Pemphigus vulgaris** (disruption of desmosomes), **Psoriasis** (accelerated cell turnover), and **Skin cancer** (basal/squamous cell carcinoma).
- **Surgical Landmarks:** **Langer's lines** (cleavage lines) are critical for minimizing scarring during incisions [Outside of source].

Embryology:

- **Origin:** Most of the skeleton arises from **mesoderm** (paraxial, lateral plate); however, most of the facial skeleton and neurocranium arise from **neural crest cells**.
- **Development:** Occurs via **intramembranous ossification** (flat bones of skull) or **endochondral ossification** (long bones).
- **Anomalies:** Includes **craniosynostosis** (premature suture fusion) and **spina bifida** (vertebral arch defect).

Histology:

- **Cell Types:** **Osteoblasts** (bone-forming), **Osteocytes** (maintenance), and **Osteoclasts** (bone-resorbing, found in Howship's lacunae).
- **Tissue Organization:** Composed of **compact bone** (outer dense layer) and **cancellous (spongy) bone** (internal trabecular network).
- **Specialized Structures:** **Osteons** (Haversian systems) are the functional units of compact bone, containing concentric lamellae around a central canal.



Integumentary System

Provides protection, regulates temperature, prevents water loss, and helps produce vitamin D. Consists of skin, hair, nails, and sweat glands.

SKELETAL SYSTEM

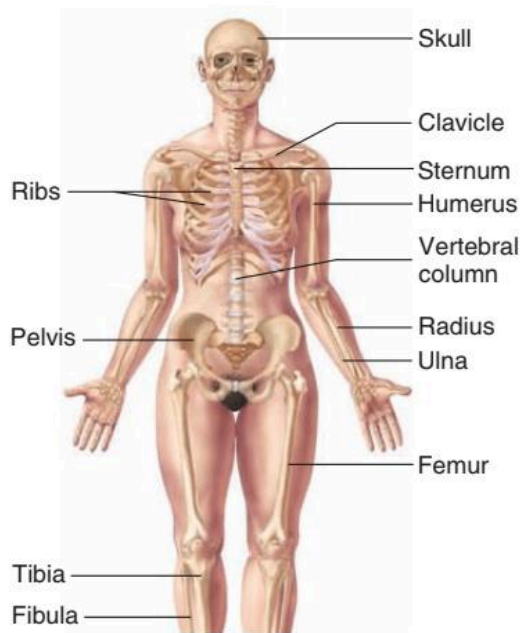
- **Functions:** Provides the body's structural framework, protects vital organs, facilitates movement through muscle attachments, and stores essential minerals like calcium and phosphorus.
- **Major Components:** Divided into the **Axial Skeleton** (80 bones: skull, vertebral column, ribs, sternum) and the **Appendicular Skeleton** (126 bones: upper and lower limbs, pectoral and pelvic girdles).

Gross Anatomy:

- **Location:** Distributed throughout the body to form the primary axis and appendages.
- **Blood Supply:** Bones are highly vascular, supplied by **nutrient arteries** that enter the diaphysis through nutrient foramina.
- **Joints:** Places where bones articulate; classified as **synarthroses** (limited/no movement) or **diarthroses** (freely mobile synovial joints).

Clinical and Imaging Anatomy:

- **Imaging:** **X-rays** are the gold standard for fractures; **DEXA scans** assess bone mineral density for osteoporosis.
- **High-Yield Correlations:** The **Pterion** is a weak point of the skull where four bones meet; a fracture here can rupture the middle meningeal artery. **Colles fracture** is a common distal radius fracture resulting in a "dinner fork" deformity.



Skeletal System

Provides protection and support, allows body movements, produces blood cells, and stores minerals and adipose. Consists of bones, associated cartilages, ligaments, and joints.

MUSCULAR SYSTEM

- **Functions:** Produces body movement, maintains posture, and generates heat through thermogenesis.
- **Major Components:** Consists of **skeletal muscle** (voluntary), **cardiac muscle** (involuntary, heart), and **smooth muscle** (involuntary, viscera and vessels).

Gross Anatomy:

- **Organization:** Muscles are grouped into functional compartments (e.g., anterior/flexor and posterior/extensor compartments of the arm) separated by intermuscular septa.
- **Innervation:** Skeletal muscles are supplied by **motor nerves** at myoneural junctions.

Embryology:

- **Origin:** Derived primarily from the **mesoderm** (specifically somites for trunk/limb muscles

and branchial arches for head/neck muscles).

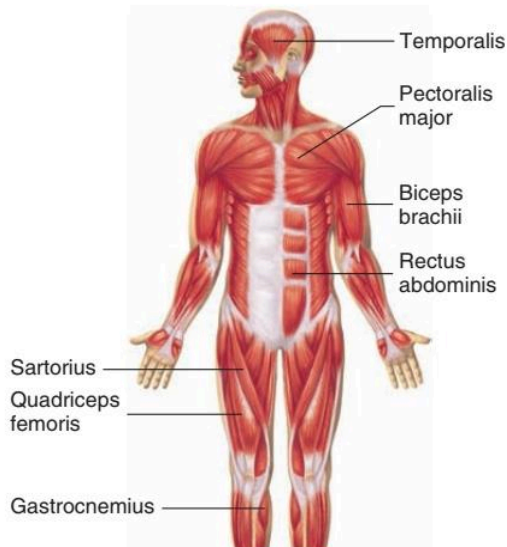
- **Development:** Myoblasts proliferate and fuse to form multinucleated **myocytes**.

Histology:

- **Functional Unit:** The **sarcomere**, defined as the segment between two **Z-discs**. It contains thick (myosin) and thin (actin) filaments.
- **Cell Types:** Elongated skeletal muscle fibers, branched cardiac myocytes with **intercalated disks**, and spindle-shaped smooth muscle cells.

Clinical Anatomy:

- **Hypertrophy:** An increase in muscle mass due to increased synthesis of contractile proteins, often resulting from resistance training.
- **Muscular Dystrophy:** Genetic disorders like **Duchenne's** involve defects in structural proteins like dystrophin.
- **High-Yield Correlations:** **Tennis elbow** (lateral epicondylitis) involves microtrauma to the common extensor origin



Muscular System

Produces body movements, maintains posture, and produces body heat. Consists of muscles attached to the skeleton by tendons.

NERVOUS SYSTEM

- **Functions:** Acts as the body's primary control and communication center; processes sensory input and initiates motor responses.
- **Major Components:** Divided into the **Central Nervous System (CNS)** (brain and spinal cord) and the **Peripheral Nervous System (PNS)** (cranial and spinal nerves).

Gross Anatomy:

- **Brain:** Composed of the cerebrum, cerebellum, and brainstem.
- **Spinal Cord:** Typically terminates at the **L1-L2 vertebral level** (conus medullaris).
- **Cranial Nerves:** 12 pairs (I-XII) that exit via various skull foramina.
- **Blood Supply:** The brain is supplied by the **internal carotid** and **vertebral arteries** (forming the Circle of Willis).

Embryology:

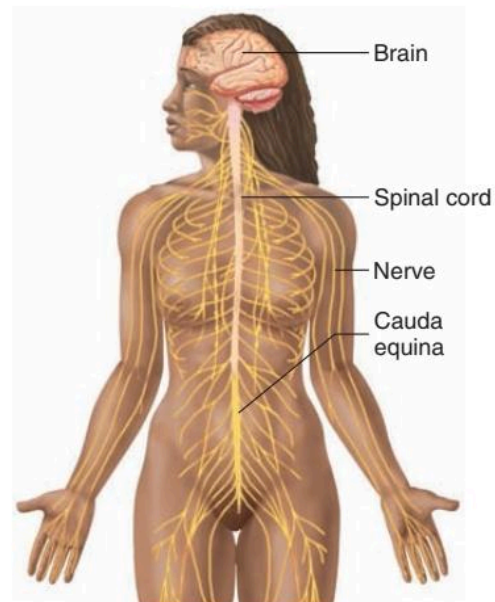
- **Origin:** Arises from **ectoderm** (specifically the neural tube for CNS and neural crest for most of PNS).

Histology:

- **Cells: Neurons** (functional signaling units) and **Glia** (support cells: astrocytes, oligodendrocytes, microglia, and Schwann cells).
- **Specialized Structures: Myelin sheaths** (formed by Schwann cells in PNS and oligodendrocytes in CNS) increase signal conduction speed via **saltatory conduction** at the nodes of Ranvier.

Clinical and Imaging Anatomy:

- **Strokes:** Focal neurological deficits usually caused by vascular occlusion in the internal carotid artery territory.
- **Shingles:** Caused by the **varicella-zoster virus** remaining dormant in the posterior root (sensory) ganglion.
- **High-Yield Correlations: A Lumbar Puncture** is safely performed below L2 (usually L4-L5) to avoid the spinal cord



Nervous System

A major regulatory system that detects sensations and controls movements, physiological processes, and intellectual functions. Consists of the brain, spinal cord, nerves, and sensory receptors.

**ENDOCRINE SYSTEM**

- **Functions:** Coordinates long-term activities of other organ systems through the secretion of **hormones** into the circulatory system.
- **Major Components:** Includes the **pituitary gland, thyroid, parathyroids, suprarenal (adrenal) glands, pancreas, pineal gland, and gonads.**

Gross Anatomy:

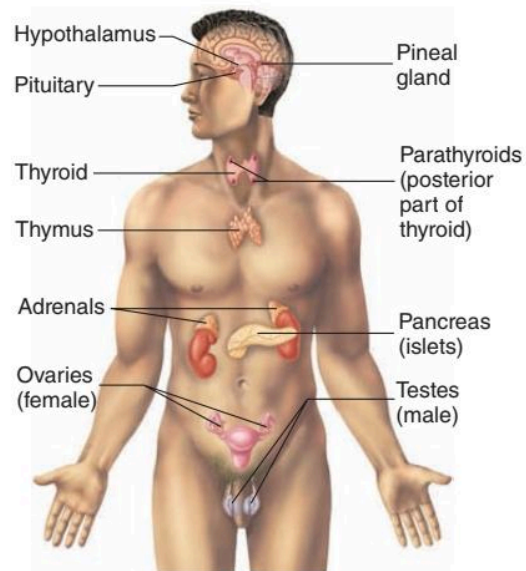
- **Pituitary Gland:** Located in the **sella turcica** of the sphenoid bone; divided into anterior and posterior lobes.
- **Thyroid Gland:** Positioned in the anterior neck, inferior to the thyroid cartilage.
- **Suprarenal Glands:** Located superior to the kidneys; consist of an outer **cortex** (steroids) and an inner **medulla** (catecholamines).

Histology:

- **Tissue Organization:** Ductless glands with a rich capillary supply to facilitate hormone distribution.
- **Suprarenal Layers:** The cortex is divided into three zones: **Zona glomerulosa** (mineraloids), **Zona fasciculata** (glucocorticoids), and **Zona reticularis** (androgens).

Clinical Anatomy:

- **Diabetes Mellitus:** Disrupted glucose homeostasis; **Type I** is insulin deficiency, while **Type II** involves insulin resistance.
- **Graves Disease:** Autoimmune hyperthyroidism resulting in nervousness, weight loss, and goiter

**Endocrine System**

A major regulatory system that influences metabolism, growth, reproduction, and many other functions. Consists of glands, such as the pituitary, that secrete hormones.

CARDIOVASCULAR SYSTEM

- **Main Functions:** A dual pump providing pulmonary (deoxygenated) and systemic (oxygenated) circulation.
- **Major Components:** Heart, arteries, veins, and capillaries.
- **Clinical Importance:** First system to function in the embryo; site of high-mortality pathologies like Myocardial Infarction (MI).

Gross Anatomy

- **Location:** Positioned in the **middle mediastinum**, surrounded by the **pericardium**.
- **Parts:** Four chambers (**2 atria, 2 ventricles**) and four valves (**tricuspid, pulmonary, mitral, aortic**).
- **Blood Supply:** **Right and Left Coronary Arteries**; the Left Anterior Descending (LAD) is the "widow-maker".
- **Venous Drainage:** Primarily via the **coronary sinus** into the right atrium.
- **Innervation:** **Sinuatrial (SA) node** (pacemaker) regulated by autonomic nerves (Vagus for parasympathetic; Sympathetic trunks).

Embryology

- **Embryonic Origin:** Derived from **splanchnic lateral plate mesoderm**.



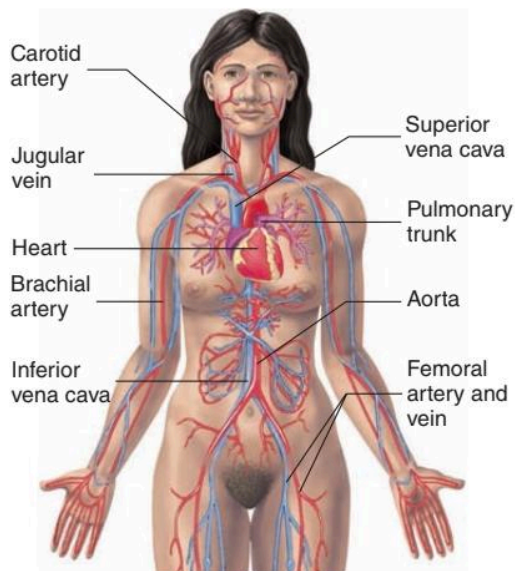
- **Development:** Heart tube forms at 3 weeks, undergoes **looping**, and partitions into four chambers by the end of the 8th week.
- **Congenital Anomalies:** **Ventricular Septal Defect (VSD)**, **Tetralogy of Fallot**, and **Patent Ductus Arteriosus (PDA)**.

Histology and Microscopic Anatomy

- **Layers:** **Endocardium** (internal lining), **Myocardium** (cardiac muscle), and **Epicardium** (visceral serous pericardium).
- **Cell Types:** Elongated, branched cardiac myocytes with **intercalated disks** containing desmosomes and gap junctions.
- **Purkinje Fibers:** Large, pale-staining modified muscle cells for rapid impulse conduction.

Imaging and Clinical Anatomy

- **Imaging:** **CT Angiogram** for coronary artery visualization; **Echocardiogram** for valve function.
- **Common Injuries:** **Cardiac Tamponade** (fluid in the pericardial sac).
- **Surgical Landmarks:** **Transverse Pericardial Sinus** used to ligate great vessels during bypass



Cardiovascular System

Transports nutrients, waste products, gases, and hormones throughout the body; plays a role in the immune response and the regulation of body temperature. Consists of the heart, blood vessels, and blood.

LYMPHATIC AND IMMUNE SYSTEM

- **Functions:** Collects excess interstitial fluid (lymph) and returns it to the venous system; serves as the site for immune surveillance and defense.
- **Major Components:** Lymphatic vessels, **lymph nodes**, **spleen**, **thymus**, and **tonsils**.

Gross Anatomy:

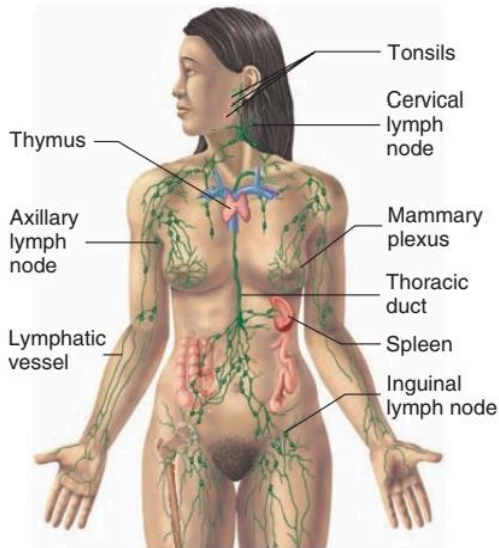
- **Drainage:** The **Thoracic Duct** drains the majority of the body into the left venous angle, while the **Right Lymphatic Duct** drains the upper right quadrant.
- **Spleen:** Located in the left upper quadrant; filters blood and removes old erythrocytes.

Histology:

- **Lymph Nodes:** Organized into a cortex (B-cell follicles) and a paracortex (T-cell zone).
- **Immune Cells:** **T-lymphocytes** (Helper CD4+ and Cytotoxic CD8+) and **B-lymphocytes** (which differentiate into antibody-secreting plasma cells).

Clinical Anatomy:

- **Sentinel Node:** The **supraclavicular (Virchow's) node** can be the first evidence of visceral malignancy.
- **Splenomegaly:** Enlargement of the spleen often seen in infections like mononucleosis



Lymphatic System

Removes foreign substances from the blood and lymph, combats disease, maintains tissue fluid balance, and absorbs dietary fats from the digestive tract. Consists of the lymphatic vessels, lymph nodes, and other lymphatic organs.

RESPIRATORY SYSTEM

- **Main Functions:** Ventilation (air movement) and gas exchange (CO_2/O_2).
- **Major Components:** **Conducting portion** (nasal cavity to terminal bronchioles) and **Respiratory portion** (respiratory bronchioles to alveoli).
- **Clinical Importance:** Vulnerable to obstruction, chronic inflammation (asthma/COPD), and malignancy.

Gross Anatomy

- **Location:** Spans head, neck, and thoracic cavity.
- **Parts:** **Trachea** (C-shaped rings), **Bronchi** (primary, lobar, segmental), and **Lungs** (3 lobes right, 2 lobes left).
- **Relations:** The **Esophagus** lies posterior to the trachea.
- **Blood Supply:** **Pulmonary arteries** (deoxygenated) for gas exchange; **Bronchial arteries** (oxygenated) for tissue nutrition.
- **Innervation:** **Vagus nerve (CN X)** (bronchoconstriction) and sympathetic trunks (bronchodilation).

Embryology

- **Embryonic Origin:** **Endoderm** (epithelial lining) and **Splanchnic Mesoderm** (cartilage and muscle).

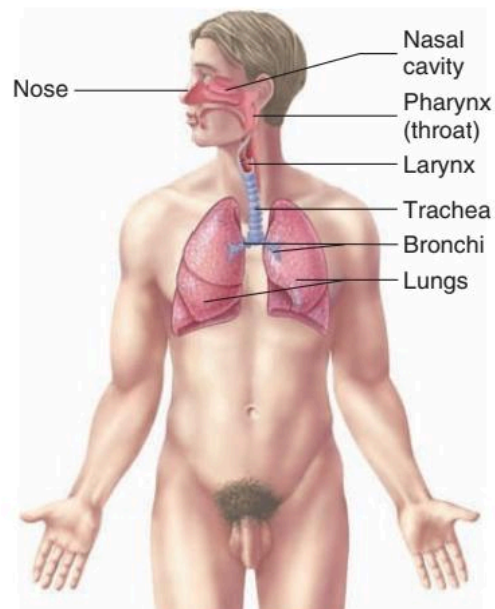
- **Development:** Begins as a **respiratory diverticulum** off the foregut during the 4th week.
- **Congenital Anomalies:** **Tracheoesophageal Fistula (TEF)** and **Infant Respiratory Distress Syndrome (IRDS)**.

Histology and Microscopic Anatomy

- **Tissue Organization:** **Respiratory epithelium** (pseudostratified ciliated columnar) with **goblet cells**.
- **Alveolar Cells:** **Type I pneumocytes** (95% surface area, gas exchange) and **Type II pneumocytes** (secrete **surfactant**).
- **Blood-Air Barrier:** Fused basal laminae of Type I pneumocytes and capillary endothelial cells.

Imaging and Clinical Anatomy

- **Imaging: Posteroanterior Radiograph** (chest X-ray) shows lung fields, heart borders, and costodiaphragmatic recesses.
- **Clinical Points: Thoracentesis** performed superior to the rib to avoid the **VAN** bundle.
- **Pneumothorax:** Entry of air into the **pleural cavity** leading to lung collapse



Respiratory System

Exchanges oxygen and carbon dioxide between the blood and air and regulates blood pH. Consists of the lungs and respiratory passages.

**DIGESTIVE SYSTEM**

- **Main Functions:** Ingestion, mechanical and chemical digestion, nutrient absorption, and elimination of solid wastes.
- **Major Components:** The alimentary canal (mouth, pharynx, esophagus, stomach, small/large intestines) and accessory organs (liver, pancreas, gallbladder, salivary glands).
- **Clinical Importance:** Site of common pathologies such as peptic ulcers, gallstones, and appendicitis.

Gross Anatomy

- **Location:** Spans from the head/neck through the thoracic and abdominopelvic cavities.
- **Blood Supply:** Primarily the **Celiac Trunk** (foregut), **Superior Mesenteric Artery (SMA)** (midgut), and **Inferior Mesenteric Artery (IMA)** (hindgut).
- **Innervation:** Autonomic supply via the **Vagus nerve (CN X)** (parasympathetic for foregut/midgut) and **Pelvic Splanchnic nerves (S2-S4)** (hindgut). Sympathetic supply via thoracic and lumbar splanchnic nerves.
- **Lymphatic Drainage:** Primarily to the celiac, superior mesenteric, and inferior mesenteric nodes.

Embryology

- **Embryonic Origin:** Epithelial lining from **endoderm**; muscular and connective tissues from **splanchnic mesoderm**.
- **Development:** The primordial gut is divided into the **foregut**, **midgut**, and **hindgut**. The midgut undergoes a 270° counterclockwise rotation and physiological herniation into the umbilical cord during week 6.
- **Congenital Anomalies:** Includes **Meckel diverticulum**, **Omphalocele**, and **Tracheoesophageal Fistula**.

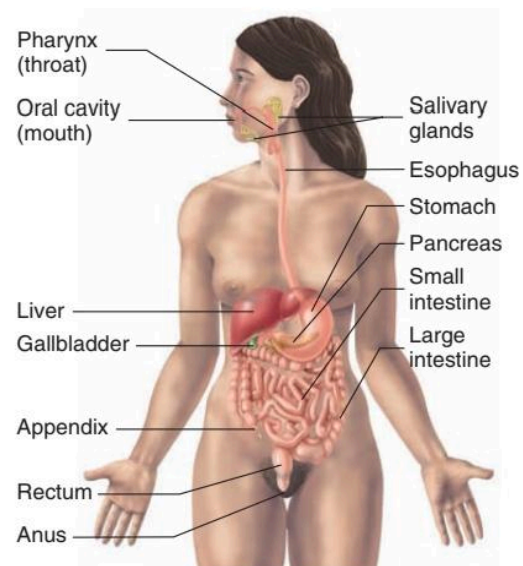
Histology

- **Tissue Organization:** Four characteristic layers: **Mucosa** (epithelium, lamina propria, muscularis mucosae), **Submucosa**, **Muscularis externa**, and **Serosa/Adventitia**.
- **Cell Types:** Includes **Oxyntic (Parietal) cells** (HCl secretion), **Chief cells** (pepsinogen), and **Goblet cells** (mucus).
- **Specialized Structures:** **Plicae circulares**, **villi**, and **microvilli** in the small intestine dramatically increase absorption surface area. **Auerbach's (myenteric)** and **Meissner's**

(submucosal) plexuses regulate motility and secretion.

Clinical Anatomy

- **McBurney's Point:** Located 1/3 of the distance from the **Right ASIS** to the umbilicus; tenderness indicates **appendicitis**.
- **Portal Hypertension:** Can lead to **caput medusae** (dilated paraumbilical veins) or esophageal varices.
- **Gallstones:** Often present with RUQ pain referred to the right shoulder due to phrenic nerve (C3-C5) irritation

**Digestive System**

Performs the mechanical and chemical processes of digestion, absorption of nutrients, and elimination of wastes. Consists of the mouth, esophagus, stomach, intestines, and accessory organs.

URINARY SYSTEM

- **Main Functions:** Filters blood to maintain electrolyte/acid-base balance and eliminates metabolic wastes (urea/creatinine).
- **Major Components:** Kidneys, ureters, urinary bladder, and urethra.
- **Clinical Importance:** Vital for blood pressure regulation via the Renin-Angiotensin-Aldosterone System (RAAS).

Gross Anatomy

- **Location:** Kidneys are **retroperitoneal** at the T12-L3 vertebral level; the right is slightly lower due to the liver.



- **Blood Supply: Renal arteries** (branches of the abdominal aorta).
- **Innervation:** Sympathetic via the **least splanchnic nerve** and celiac/aorticorenal ganglia; parasympathetic via the vagus nerve and pelvic splanchnics.

Embryology

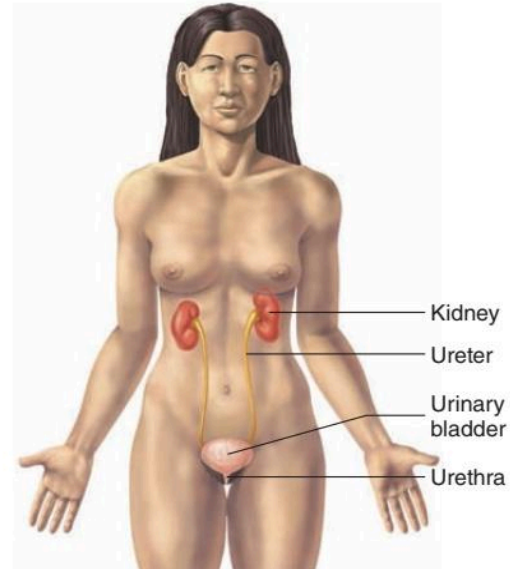
- **Embryonic Origin:** Derived from the **intermediate mesoderm**.
- **Development:** Three sets of kidneys form: the pronephros (vestigial), mesonephros (interim), and **metanephros** (permanent kidney).
- **Congenital Anomalies:** **Renal agenesis**, **Ectopic kidneys**, and **Patent urachus**.

Histology

- **Functional Unit:** The **Nephron**, consisting of the renal corpuscle (glomerulus and Bowman's capsule) and the tubule system.
- **Specialized Structures:** **Podocytes** form the filtration slits; the **Juxtaglomerular Apparatus (JGA)** monitors blood pressure and sodium.
- **Ureters/Bladder:** Lined with **transitional epithelium (urothelium)** to allow for stretching.

Clinical Anatomy

- **Kidney Stones (Calculi):** Can cause intense pain ("renal colic") typically referred to the T11-L2 dermatomes (loin to groin).
- **Hydronephrosis:** Dilation of the renal pelvis due to downstream obstruction.



Urinary System

Removes waste products from the blood and regulates blood pH, ion balance, and water balance. Consists of the kidneys, urinary bladder, and ducts that carry urine.

REPRODUCTIVE SYSTEM

- **Main Functions:** Production of gametes (sperm/oocytes) and sex hormones, and support of the developing fetus (female).
- **Clinical Importance:** High-yield for embryological development, infertility workups, and oncology (prostate/ovarian cancer).

Gross Anatomy (Male/Female)

- **Gonads: Testes** (located in the scrotum) and **Ovaries** (located in the lateral pelvic wall).
- **Ducts: Vas deferens** and **Epididymis** (male); **Uterine (Fallopian) tubes** and **Uterus** (female).
- **External Genitalia:** Penis and Scrotum; Vulva (Labia, Clitoris).
- **Blood Supply: Gonadal (Testicular/Ovarian) arteries** from the abdominal aorta. **Internal pudendal artery** for external genitalia.

Embryology

- **Sex Determination:** Dependent on the **SRY gene** on the Y chromosome, which induces testes differentiation.
- **Duct Development: Mesonephric (Wolffian) ducts** form the male reproductive tract; **Paramesonephric (Müllerian) ducts** form the female tract.

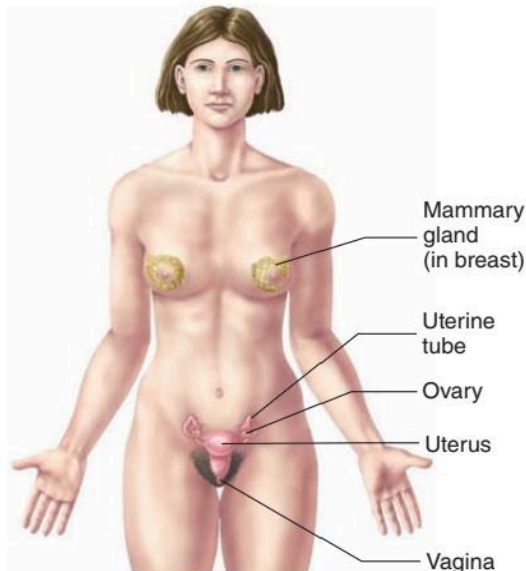
Histology



- **Male: Seminiferous tubules** contain **Sertoli cells** (support) and **Spermatogenic cells**. **Leydig cells** produce testosterone.
- **Female: Ovarian follicles** at various stages (primordial, primary, secondary, antral, Graafian). The **Corpus Luteum** produces progesterone after ovulation.

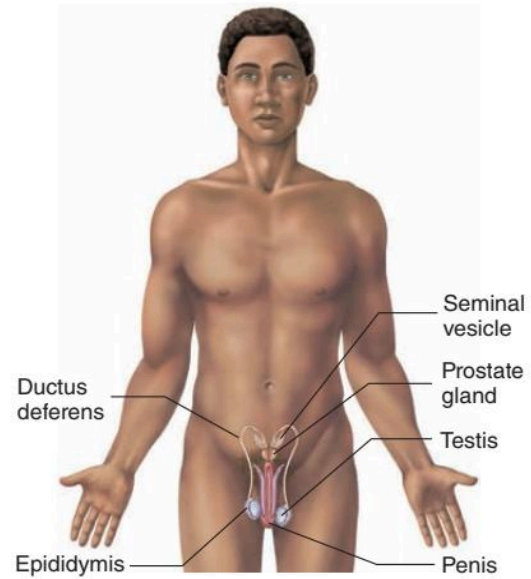
Clinical Anatomy

- **Ectopic Pregnancy:** Most commonly occurs in the **ampulla** of the uterine tube.
- **Varicocele:** A "bag of worms" appearance in the scrotum due to pampiniform plexus dilation, often on the left side.
- **Erection/Ejaculation:** Erection is parasympathetic (**Point = Parasympathetic**); ejaculation is sympathetic (**Shoot = Sympathetic**)



Female Reproductive System

Produces oocytes and is the site of fertilization and fetal development; produces milk for the newborn; produces hormones that influence sexual function and behaviors. Consists of the ovaries, uterine tubes, uterus, vagina, mammary glands, and associated structures.



Male Reproductive System

Produces and transfers sperm cells to the female and produces hormones that influence sexual functions and behaviors. Consists of the testes, accessory structures, ducts, and penis.